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Classifying Amphiprion Ocellaris Using a Custom CNN Architecture: A Comparative Study with a pre-build CNN for Ecological Monitoring

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Abstract
Classification of marine species like *Amphiprion ocellaris* is vital for ecological monitoring and conservation. This study uses a convolutional neural network (CNN) and a pre-trained MobileNetV2 model to classify *A. ocellaris* from a dataset of 1130 images sourced from the GBIF library, including related species and corals. A simple CNN architecture was developed to address challenges of visual similarity among species and background elements. The CNN achieved a precision of 97.56% and a recall of 100%, making it effective for identifying target species. While the CNN slightly outperformed MobileNetV2 in recall, MobileNetV2 demonstrated comparable precision and offered practical advantages in terms of time efficiency and ease of implementation. This research highlights the value of deep learning models in ecological studies, emphasizing transfer learning for its flexibility and potential for generalization. Both approaches enable efficient filtering of large datasets to identify species like *A. ocellaris*, providing a solid foundation for biodiversity monitoring efforts.

1. Introduction

The false clownfish, *Amphiprion ocellaris*, is a species of significant ecological interest due to its role within the damselfish family. These fishes possess unique traits, such as their distinct coloration used for intra-specific communication [1], including territory display, conspecific recognition, and reproductive maturity indicators. Their ecological importance is further highlighted by their radiation, allowing them to colonize coral reefs and develop symbiotic relationships with sea anemones [3]. However, distinguishing *A. ocellaris* from visually similar species remains challenging, especially when performed manually [2].

Manual identification, while common, is time-consuming and prone to errors, particularly with large datasets. To address this, our research explores the use of convolutional neural networks (CNNs), a deep learning technique well-suited for image classification tasks. Specifically, we developed a simple CNN architecture to determine whether an image contains *A. ocellaris*, demonstrating that a cost-effective, easy-to-implement model can achieve high performance. Such a tool holds great potential for laboratory and field use, enabling efficient filtering of large datasets and quick identification of *A. ocellaris*.

Furthermore, to evaluate our approach, we compared the performance of the custom-built CNN against a pre-trained MobileNetV2 model fine-tuned on the same dataset. This comparison aimed to identify the optimal method for recognizing *A. ocellaris*: whether a simple, tailored architecture is sufficient or if transfer learning with a pre-built model provides better results. The images were sourced from GBIF, an open-access biodiversity repository, ensuring diversity and applicability to real-world ecological studies.

2. Data

The dataset used in this study was sourced entirely from GBIF, an open-access biodiversity repository. The images are RGB photos of *Amphiprion ocellaris* (false clownfish) in their natural environment, alongside several other marine species. Specifically, the dataset includes 500 images of *A. ocellaris* and 630 images of other species, such as *Amphiprion clarkii*, *Neoglyphidodon oxyodon*, *Neopetrolisthes maculatus*, and *Heteractis aurora*. These images are organized into two categories: target (*A. ocellaris*) and non-target (other species).

Figure 1. *Subset of the dataset displaying examples of marine species. From top-left to bottom-right: Amphiprion ocellaris, Amphiprion clarkii, Amphiprion ocellaris, Neopetrolisthes maculatus, Neoglyphidodon oxyodon, and Neopetrolisthes maculatus on Heteractis aurora .*

2.1. Splitting Data

The dataset was split into training, validation, and testing sets using a function that divided the data into train, validation, and test folders with respective ratios of 70%, 15%, and 15%. This resulted in training data consisting of 719 images, validation data of 205 images, and testing data of 203 images.

2.2. Pre-processing

To address the inherent class imbalance, weighting was applied to ensure a balanced representation of both target and non-target categories during training with weights of 1.23 and 0.77 assigned to each class respectively. The images were resized to a resolution of 224 x 224 pixels using the default interpolation method of the Python Imaging Library (PIL). Subsequently, all pixel values were normalized to a range of 0 to 1 by dividing by 255. During the dataset preparation, manual inspection was performed to discard very low-quality images, ensuring only usable images were retained. Additionally, any corrupted or missing images encountered in the dataset were removed. Class labels were encoded as binary values, with 0 representing non-target species and 1 representing *A. ocellaris*. This simple binary encoding allowed for straightforward classification of the images.

2.3. Data Augmentation

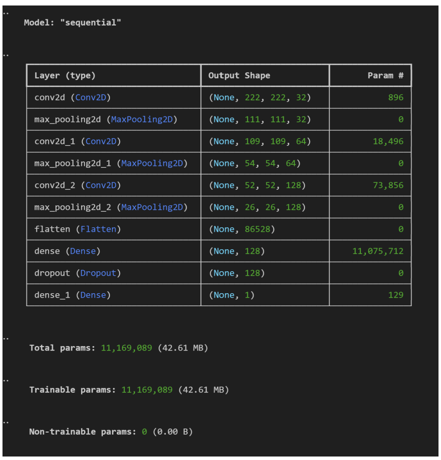
Data augmentation was performed using the ImageDataGenerator utility from the open-source machine learning library TensorFlow (www.tensorflow.org; version 2.18.0). The training set was expanded to 4620 images.

3. Model

3.1. CNN Architecture

The convolutional neural network (CNN) model used in this study was implemented using TensorFlow and Keras. The architecture is a Sequential model consisting of the following layers:

- **Convolutional Layers**: Three convolutional layers were used, with 32, 64, and 128 filters, respectively, and a kernel size of (3, 3). Each layer used the ReLU activation function to introduce non-linearity.
- **Pooling Layers**: Each convolutional layer is followed by a max pooling layer with a pool size of (2, 2), which helps reduce the spatial dimensions and computational complexity.
- **Flatten Layer**: After the final convolutional layer, the output is flattened to a 1D vector to prepare for the fully connected layers.
- **Fully Connected Layer**: A dense layer with 128 units and ReLU activation was used.
- **Dropout Layer**: A dropout layer with a rate of 0.5 was added to prevent overfitting by randomly setting half of the units to zero during training.
- **Output Layer**: The final dense layer has 1 unit with a sigmoid activation function for binary classification (target vs. non-target).



Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 222, 222, 32)	896
max_pooling2d (MaxPooling2D)	(None, 111, 111, 32)	0
conv2d_1 (Conv2D)	(None, 109, 109, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 54, 54, 64)	0
conv2d_2 (Conv2D)	(None, 52, 52, 128)	73,856
max_pooling2d_2 (MaxPooling2D)	(None, 26, 26, 128)	0
flatten (Flatten)	(None, 86528)	0
dense (Dense)	(None, 128)	11,075,712
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 1)	129

Total params: 11,169,089 (42.61 MB)

Trainable params: 11,169,089 (42.61 MB)

Non-trainable params: 0 (0.00 B)

Figure 2. *Architecture of the custom built CNN with its hyper-parameters.*

The model was compiled using the Adam optimizer, with a learning rate dynamically adjusted using early stopping and a learning rate scheduler from the TensorFlow library. In addition to the custom CNN, a MobileNetV2 model was used for comparison through transfer learning. The MobileNetV2 architecture was fine-tuned on the same dataset, with similar optimization strategies applied to ensure a fair performance evaluation.

4. Results

The model was evaluated on the test set, yielding the following performance metrics:

- **Accuracy**: 99.02%
- **Precision**: 97.56%
- **Recall**: 100%
- **AUC**: 98.00%

In comparison, a MobileNetV2 model achieved an accuracy of 97.12%, a recall of 97.12%, and a precision of 97.15%. While the custom-built CNN showed slightly higher metrics in some areas, both models performed similarly well, indicating that MobileNetV2 is also a highly reliable model for this classification task.

5. Performance Evaluation

Figure 3 shows the accuracy over the epochs for the custom CNN, while Figure 4 shows the evolution of the training and validation loss over the epochs. The custom model demonstrated good convergence behavior, with both training and validation accuracy increasing while the loss decreased. Similar results were obtained with the MobileNetV2 model, showing comparable trends in training and validation metrics over the epochs.

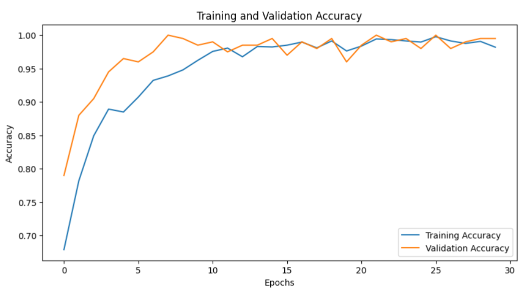


Figure 3. *Training and validation accuracy over the epochs.*

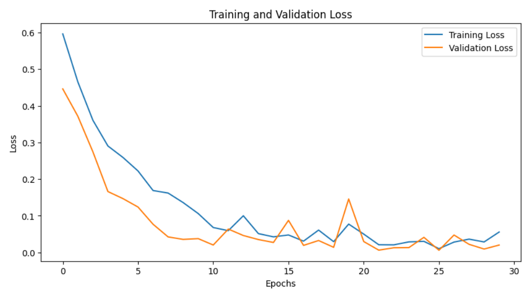


Figure 4. *Training and validation loss over the epochs.*

The confusion matrices provide further insight into the performance of both models. Figure 5 and 6 show that the custom-built CNN and MobileNetV2 models achieve a high true positive rate, effectively identifying images containing *A. ocellaris*. Both models demonstrate a low number of false positives and false negatives, highlighting their reliability in distinguishing between target and non-target classes.

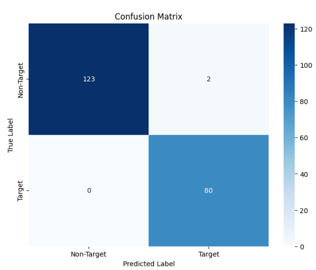


Figure 5. *Confusion matrix for the custom-built CNN.*

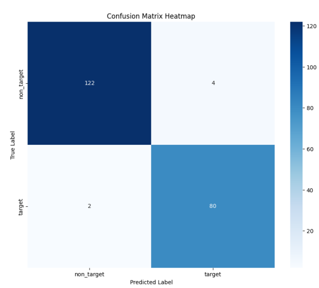


Figure 6. *Confusion matrix for MobileNetV2.*

6. XAI Clarification

To better understand how the model makes decisions, a saliency map was generated using gradient-based visualization. Initially, the model appeared to focus on the sea anemone when identifying *A. ocellaris*, suggesting it was using environmental context rather than directly identifying the fish. After augmenting the dataset with additional samples of anemone in the non-target class, the model shifted its focus to the distinctive shape and pattern of the fish itself, as shown in Figure 7. This improvement indicates that the model now primarily relies on the fish's body pattern and shape for classification, suggesting it has learned to accurately distinguish *A. ocellaris* based on unique visual features instead of relying on contextual clues. A saliency map of the transfer learning model (MobileNetV2) was also generated, and it revealed that the model tends to focus more broadly across the entire image rather than specific features.

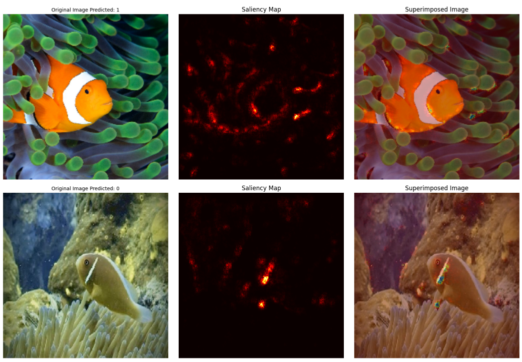


Figure 7. *Original image, saliency map, and superimposed image for the custom-built CNN highlighting regions influencing classification.*

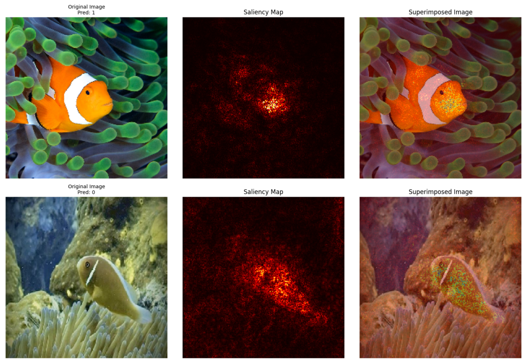


Figure 8. *Original image, saliency map, and superimposed image for MobileNetV2 highlighting regions influencing classification.*

7. Discussion

The CNN and MobileNetV2 models both delivered strong results in classifying images of *Amphiprion ocellaris* from other marine species. The CNN achieved a slightly better recall, but since precision is the priority for filtering data and confidently isolating target images, both models proved effective for this task.

The accuracy and loss over epochs revealed no signs of overfitting, confirming that both models were trained effectively. The saliency maps added valuable insights into how the models make decisions. The CNN concentrated on specific patterns, while the MobileNetV2 model adopted a broader focus across the images, reflecting its pre-trained architecture. These differences help explain the slight variations in recall and precision between the models.

Transfer learning with MobileNetV2 stands out for its practical benefits. It saves time by enabling a focus on critical tasks like fine-tuning and dataset improvements rather than starting from scratch. Additionally, MobileNetV2 likely generalizes better to new data, although this requires further testing.

To conclude, both methods are solid options. However, we recommend transfer learning for the clear advantages it offers in scenarios where precision and ease of implementation are essential.

8. Resources

The dataset used in this study can be accessed at the following link: https://drive.google.com/drive/folders/1_RLQ1ae4UUV_Ohlrczp7pvl8OYsj2R6a?usp=sharing. The code developed for training and evaluation analysis, for both the custom CNN and MobileNetV2 models, is available at: https://github.com/Ayoub-Hartaoui/MLEES_Portfolio.

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